Tuning Parameter Reference & Implementation Guide: LQR, Fixed Super-Twisting SMC, and Adaptive Super-Twisting SMC

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I. EXECUTIVE TUNING PARAMETER MATRIX

Control Method	Primary Parameters	Typical Value Range	Systematic Tuning Criteria	Main Advantages / Trade-Offs
1. Linear Quadratic Regulator (LQR)	**Q**: State penalty **R**: Control effort penalty	Q = diag([100, 1, 100, 1]) R = 0.01 to 10.0	**Bryson's Rule:** $Q_{ii} = 1 / (x_{i,max})^2$ $R_{jj} = \rho / (u_{j,max})^2$	Guaranteed gain margins (∞) & phase margin (≥60°). Optimal for linear models; degraded by unmodeled nonlinearities and saturations.
2. Fixed Super-Twisting SMC (STSMC)	λ: Sliding slope $\mathbf{k_1}$: Fractional gain $\mathbf{k_2}$: Integral gain	λ = 5 to 30 $k_1 = 1.5\sqrt{L} \approx 3$ to 15 $k_2 = 1.1\ L \approx 10$ to 50	**Lyapunov Criterion:** $k_1 = 1.5\sqrt{L}$ $k_2 = 1.1\ L$ where $\lvert d'(t)\rvert \le L$	Finite-time convergence of s and ds/dt to zero. Eliminates high-frequency chattering without low-pass filtering. Requires known upper disturbance bound L.
3. Adaptive Super-Twisting SMC	α_{min}: Base gain ρ: Adaptation rate μ: Deadband width σ: Leakage rate	α_{min} = 1.0 to 5.0 ρ = 5.0 to 20.0 μ = 0.005 to 0.05 σ = 0.5 to 2.0	**Anti-Windup Update:** $d\alpha/dt = \rho\sqrt{\lvert s\rvert}$ for $\lvert s\rvert > \mu$ $d\alpha/dt = -\sigma(\alpha - \alpha_{min})$ for $\lvert s\rvert \le \mu$ $\beta(t) = 2\ \alpha(t)$	Dynamically adjusts gains online to reject unknown, time-varying disturbances without knowing L. Deadband prevents gain drift and eliminates chatter.

II. LINEAR QUADRATIC REGULATOR (LQR) FORMULATION & TUNING

1. Plant Representation: For a continuous-time linear state-space model $x'(t) = A\,x(t) + B\,u(t)$ with full state vector $x \in \Re^n$ and control input $u \in \Re^m$.

2. Control Law & Optimization: The optimal state feedback is $\mathbf{u = -K\,x}$, which minimizes the quadratic cost functional:

$$J = \int_0^{\infty} (x^T Q\, x + u^T R\, u)\, dt$$

The optimal feedback gain matrix is computed as $\mathbf{K = R^{-1}\,B^T\,P}$, where $P = P^T > 0$ is the unique positive-definite solution to the continuous-time **Algebraic Riccati Equation (ARE)**:

$$A^T P + P A - P B R^{-1} B^T P + Q = 0$$

3. Tuning Methodology & Physical Trade-Offs:

• **State Weighting Matrix Q ≥ 0:** Penalizes deviations of state variables from zero. Increasing Q_{ii} accelerates the decay rate of $x_i(t)$, yielding faster transient rise times and lower steady-state tracking errors.

• **Control Weighting Matrix R > 0:** Penalizes control actuator effort. Increasing R_{jj} prevents control saturation, reduces high-frequency actuator wear, and damps excessive voltage/current spikes, at the expense of slower closed-loop dynamics.

• **Bryson's Tuning Rule:** Set $Q_{ii} = 1 / (x_{i,max})^2$ where $x_{i,max}$ is the maximum permissible excursion of state i, and $R_{jj} = \rho / (u_{j,max})^2$ where $u_{j,max}$ is actuator saturation limit and $\rho \in (0, 1]$ is an aggressiveness factor.

Parameter	Symbol	Default Value	Tuning Action / Guidelines
State Matrix Weight	Q	diag([100, 1, 100, 1])	Increase diagonal elements corresponding to priority states (e.g. angle or position error) to enforce tighter tracking.
Effort Penalty Weight	R	0.1 to 1.0	Decrease R for faster, more aggressive disturbance rejection; increase R if actuator hits hard torque/voltage saturation limits.
Cross-Term Weight	N	0 (zero matrix)	Set to zero for uncoupled state-control penalties; only non-zero when state and control cross-coupling explicitly exists.

III. FIXED SUPER-TWISTING SLIDING MODE CONTROL (FIXED STSMC)

1. Concept & Chattering Elimination: Standard First-Order Sliding Mode Control enforces $s = 0$ by applying high-frequency switching $u = -K\,sign(s)$, causing severe mechanical vibration and chattering. The **Super-Twisting Algorithm** (Levant, 1993) is a 2nd-order sliding mode technique that drives both $s(t) = 0$ and $s'(t) = 0$ in finite time by embedding the discontinuous signum function beneath a continuous integrator, producing a strictly continuous control signal.

2. Sliding Surface Definition: For tracking error $e(t) = x(t) - x_d(t)$, define the sliding manifold:
$\mathbf{s(t) = e'(t) + \lambda\, e(t)}$ (where $\lambda > 0$ dictates error decay bandwidth once on the sliding surface)

3. Super-Twisting Control Law: The control signal comprises an equivalent feedforward term u_{eq} and super-twisting term u_{st}:
$\mathbf{u(t) = u_{eq}(t) - k_1\,|s|^{1/2}\,sign(s) + v(t)}$
$\mathbf{v'(t) = -k_2\,sign(s)}$

4. Lyapunov Stability & Finite-Time Convergence Criterion:
Assume the time derivative of the external disturbance is bounded by $|d'(t)| \le L$. Based on the strict Lyapunov matrix inequality derived by Moreno & Osorio (2012), global finite-time convergence to the origin is guaranteed if gains are chosen as:
$\mathbf{k_1 = 1.5\,\sqrt{L}}$, $\mathbf{k_2 = 1.1\,L}$

Parameter	Symbol	Typical Value	Physical Role & Tuning Guidelines
Sliding Slope	λ	5.0 to 25.0	Governs error convergence speed on the manifold: e(t) = e(0) exp(-λ t). Higher λ increases tracking speed but requires higher control authority.
Fractional Gain	k_1	$1.5\sqrt{L} \approx 3.0$ to 12.0	Proportional fractional switching term. Dominates transient convergence toward the sliding surface |s| = 0.
Integral Gain	k_2	$1.1\,L \approx 10.0$ to 45.0	Integral switching term. Cancels smooth, unknown matched disturbances and parameter perturbations in steady state.
Boundary Smoothing	ε_{st}	10^{-4} to 10^{-3}	Smooths discontinuous sign(s) into s / (|s| + ε_{st}) in digital implementations to eliminate numerical chatter.

IV. ADAPTIVE SUPER-TWISTING SMC (ADAPTIVE STSMC)

1. Problem with Fixed Gains: In practical environments, the disturbance bound L is rarely known a priori and varies with operating conditions. Overestimating L leads to excessively large fixed gains k_1, k_2, causing control saturation and high-frequency chatter. Underestimating L causes loss of sliding motion.

2. Adaptive Control Structure: The adaptive super-twisting control law replaces fixed gains with time-varying gains $\alpha(t)$ and $\beta(t)$:
$\mathbf{u(t) = u_{eq}(t) - \alpha(t)\,|s|^{1/2}\,sign(s) + v(t)}$
$\mathbf{v'(t) = -\beta(t)\,sign(s)}$, where $\mathbf{\beta(t) = \gamma\,\alpha(t)}$ (or $\beta = \alpha^2 / 4$)

3. Anti-Windup Adaptive Update Law with Deadband & Leakage:
To prevent unbounded gain inflation due to unmodeled high-frequency sensor noise, gains are updated with a deadband threshold μ and linear leakage rate σ:
$\mathbf{\alpha'(t) = \rho\,|s|^{1/2}}$ *if* $|s(t)| > \mu$ (Gain ramps up aggressively to eliminate tracking error)
$\mathbf{\alpha'(t) = -\sigma\,(\alpha(t) - \alpha_{min})}$ *if* $|s(t)| \le \mu$ (Gain smoothly decays toward minimum baseline to suppress chattering)

Parameter	Symbol	Typical Value	Physical Role & Tuning Guidelines
Minimum Base Gain	α_{min}	1.5 to 3.0	Lower bound for proportional gain. Guarantees nominal control authority when system is inside the deadband.
Adaptation Rate	ρ	5.0 to 15.0	Rate of gain increase. Set higher if system experiences sharp, fast disturbances; set lower to prevent gain overshoot.
Deadband Width	μ	0.005 to 0.02	Threshold boundary. Defines the sliding precision. Must be set larger than sensor noise amplitude to prevent drift.
Leakage Decay Rate	σ	0.8 to 2.0	Rate at which gain decays back to α_{min} when inside deadband. Eliminates steady-state chattering.
Coupling Gain Ratio	γ	1.5 to 2.5	Proportional-to-integral gain multiplier. Standard value $\gamma = 2.0$ satisfies the Shtessel-Edwards Lyapunov condition.

V. SIMULINK ARCHITECTURE & WIRING GUIDE

• **LQR Block:** Feed state vector $x = [pos, vel, theta, theta_dot]^T$ into a Gain block set to -K_lqr (Multiplication: *Matrix K*u*).
• **Fixed STSMC Block:** Compute s = e_dot + lambda_smc * e. Pass s into a MATLAB Function block executing: u_st = -k1 * sqrt(abs(s)) * (s/(abs(s)+eps)) + v, with v fed from an Integrator block with derivative v_dot = -k2 * (s/(abs(s)+eps)).
• **Adaptive STSMC Block:** Integrate alpha from initial condition alpha_min. Compute alpha_dot = (abs(s) > mu) * (rho * sqrt(abs(s))) - (abs(s) <= mu) * (sigma * (alpha - alpha_min)), and assign beta = gamma * alpha.
• **Solver Setup:** Use fixed-step solver: ode4 (Runge-Kutta) with fixed step size Δt = 1e-4 s to eliminate numerical chattering.

VI. COMPLETE READY-TO-RUN MATLAB INITIALIZATION SCRIPT (init_controllers.m)

```
%% MATLAB / SIMULINK INITIALIZATION SCRIPT: LQR vs FIXED STSMC vs ADAPTIVE STSMC
clear; clc;
% 1. PLANT MATRICES (Example: Linearized Inverted Pendulum on a Cart)
M = 1.0; m = 0.1; l = 0.5; g = 9.81;
A = [0, 1, 0, 0; 0, 0, -m*g/M, 0; 0, 0, 0, 1; 0, 0, (M+m)*g/(M*l), 0];
B = [0; 1/M; 0; -1/(M*l)]; C = eye(4); D = zeros(4,1);

% 2. LQR CONTROLLER PARAMETERS
Q_lqr = diag([100.0, 1.0, 200.0, 1.0]); % State penalty (position & angle)
R_lqr = 0.1;                            % Control effort penalty
[K_lqr, P_lqr, E_lqr] = lqr(A, B, Q_lqr, R_lqr);
fprintf('[1] LQR Gain Matrix K: '); disp(K_lqr);

% 3. FIXED SUPER-TWISTING SMC (STSMC) PARAMETERS
lambda_smc  = 12.0;    % Sliding manifold slope: s = e_dot + lambda*e
L_dist_est  = 15.0;    % Assumed upper bound on disturbance derivative: |d_dot| <= L
k1_fixed    = 1.5 * sqrt(L_dist_est);  % Proportional fractional gain (~5.81)
k2_fixed    = 1.1 * L_dist_est;         % Integral switching gain (~16.50)
eps_smooth  = 1e-4;                     % Boundary layer smoothing: s / (|s| + eps)
fprintf('[2] Fixed STSMC Initialized: lambda=%.1f, k1=%.2f, k2=%.2f\n', lambda_smc, k1_fixed, k2_fixed);

% 4. ADAPTIVE SUPER-TWISTING SMC (ASTSMC) PARAMETERS
alpha_min   = 2.0;     % Minimum baseline proportional gain
alpha_init  = 2.5;     % Initial gain at t = 0
rho_adapt   = 10.0;    % Adaptation acceleration rate (gain increase when |s| > mu)
mu_deadband = 0.01;    % Deadband threshold (|s| <= mu triggers gain decay)
sigma_leak  = 1.2;     % Gain leakage/decay rate to prevent drift
gamma_rel   = 2.0;     % Proportional-to-integral ratio: beta(t) = gamma * alpha(t)
fprintf('[3] Adaptive STSMC Initialized: alpha_min=%.1f, rho=%.1f, mu=%.3f, sigma=%.1f\n', ...
    alpha_min, rho_adapt, mu_deadband, sigma_leak);
fprintf('================================================================================\n');
fprintf(' Workspace ready! You can now launch your Simulink simulation.\n');
```

VII. TROUBLESHOOTING & COMMON PITFALLS

1. Chattering in Fixed STSMC: If micro-chatter persists, increase the smoothing factor ε_{st} from 1e-4 to 1e-3, or reduce solver step size Δt from 1e-3 s to 1e-4 s.
2. Gain Inflation / Windup in Adaptive STSMC: If $\alpha(t)$ drifts to excessively large values during steady-state operation, the deadband μ is set too small relative to measurement noise. Increase μ (e.g. from 0.005 to 0.02) or increase leakage rate σ.
3. Algebraic Loops in Simulink: Ensure that the integrator state for $v(t)$ in STSMC does not form a direct algebraic loop with the plant input without passing through an explicit integration or unit delay block.